

# **Comparative Analysis of Machine Learning Classifiers for Multilabel Classification Problem**

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## **Abstract**

Multilabel classification is a machine learning problem in which each instance may belong to multiple classes simultaneously. Unlike multiclass classification where only one class label is assigned to each sample, multilabel learning must capture label correlations and dependencies among outputs. This study presents a comparative analysis of several multilabel classification techniques applied on diverse datasets from domains such as medical imaging, retail analytics, network analysis, and image recognition. The evaluated approaches include Binary Relevance, Classifier Chains, Random k-Labelsets (RAkEL) and multilabel Decision Trees. Performance is evaluated using multilabel-specific metrics including Hamming Loss, Subset Accuracy, Micro and Macro averaged F1-score and ROC-AUC.

## **1. Introduction**

Multilabel classification is an important supervised learning problem in which a single instance may simultaneously belong to multiple predefined categories. Unlike traditional multiclass classification where exactly one class label is assigned to each sample, multilabel problems require predicting a subset of labels for every instance. This increases the complexity of the learning task due to label dependencies, high-dimensional output spaces, and imbalanced label distributions.

Despite its broad applicability, multilabel classification introduces several challenges. Traditional machine learning algorithms are primarily designed for single-label tasks and therefore require adaptation for multilabel prediction. Additionally, multilabel datasets often exhibit sparse label distributions, label imbalance, and strong inter-label correlations which significantly influence classifier performance.

To address these challenges, several multilabel learning strategies have been proposed. Problem transformation methods such as Binary Relevance, Classifier Chains, and Label Powerset convert multilabel tasks into simpler learning problems. Ensemble-based methods such as Random k-Labelsets (RAkEL) attempt to model label correlations more effectively. Algorithm adaptation approaches including ML-kNN and multilabel Decision Trees directly extend conventional algorithms to support multilabel outputs.

In this study, a comprehensive comparative analysis of multiple multilabel classification techniques is conducted across several datasets from diverse domains. Various multilabel classifiers are evaluated using consistent preprocessing, validation strategies, and evaluation metrics. The primary objective is to analyze how different multilabel approaches perform under varying dataset characteristics and to identify suitable methods for practical multilabel learning problems.

### 3.1. Datasets

The study utilizes multiple datasets from enzyme substrate detection and drug consumption:

1. *Drug Consumption Classification Dataset [1]:*

*Dataset contains behavioural information with demographics used to classify the person as a user or non-user of a particular drug. There are several drugs in the dataset. We have used Nicotine, Cannabis, Ecstasy and Legalh for our predictions: 4 labels.*

2. *Enzyme Substrate with Chemical Information [2]:*

*Enzymes are known to act on molecules with structural similarities with their substrates. This behaviour is called promiscuity. Scientists working in drug discovery use this behaviour to target/design drugs to either block or promote biological actions. In this dataset, we have used Chemical Information data to detect for the Enzyme substrate.*

*We have 6 labels: EC1, EC2, EC3, EC4, EC5, EC6*

3. *Enzyme Substrate with Structural Information [2]:*

*With the similar problem statement as Dataset 2, we use Structural information to predict the enzyme substrate.*

4. *Enzyme Substrate with Connectivity Information [2]:*

*Similar problem statement as Dataset 2, however we use connectivity information to predict the enzyme substrate.*

The table below contains information about the shape of dataset:

| <b>S.no</b> | <b>#Instances</b> | <b>#Features</b> | <b>#Labels</b> |
|-------------|-------------------|------------------|----------------|
| 1           | 1885              | 16               | 4              |
| 2           | 1089              | 196              | 6              |
| 3           | 1039              | 511              | 6              |
| 4           | 1039              | 511              | 6              |

### 3.2. Preprocessing

Prior to model evaluation, all datasets undergo preprocessing to ensure consistency and compatibility with multilabel learning algorithms.

- Missing values are removed
- Categorical variables are encoded using one-hot encoding techniques.
- In the Drug Classification Dataset, each target variable was initially multiclass which is binarized by taking the “Never Used”, “Used over a decade” and “Used in the Last Decade” category as the “Non-Users” while the rest i.e., “Used in the past year/month/week/day” as the “Users”.
- For the Enzyme Substrate Datasets, the target variables are transformed into binary multilabel vectors using one-hot encoding scheme.

### 3.3. Classification Models

Several multilabel classification techniques are evaluated in this study.

#### Binary Relevance (BR)

Binary Relevance transforms the multilabel problem into multiple independent binary classification tasks, one for each label. Logistic Regression classifiers are used as the base learners.

#### Classifier Chains (CC)

Classifier Chains model label dependencies by sequentially passing predictions from previous classifiers as additional input features to subsequent classifiers.

#### Random k-Labelsets (RAkEL)

RAkEL is an ensemble-based multilabel technique that trains classifiers on random subsets of labels. It effectively captures label correlations while reducing the complexity associated with Label Powerset methods.

#### Multilabel Decision Tree

Decision Trees inherently support multilabel outputs by recursively partitioning the feature space based on multilabel impurity criteria.

### 3.4. Evaluation Metrics

Model performance is evaluated using several multilabel-specific evaluation metrics.

**Subset Accuracy:** Subset Accuracy measures the proportion of instances for which the predicted label set exactly matches the true label set.

**Hamming Loss:** Hamming Loss computes the fraction of incorrectly predicted labels to the total number of labels. Lower values indicate better performance.

**F1-Score:** The F1-score is the harmonic mean of Precision and Recall and provides a balanced measure of multilabel classification performance.

Both Micro and Macro averaged F1-scores are evaluated:

- **Micro averaging** aggregates contributions across all labels.
- **Macro averaging** computes metrics independently for each label and averages them.

**ROC-AUC:** Receiver Operating Characteristic Area Under Curve evaluates ranking quality and probabilistic discrimination performance across labels.

For validation, Holdout method with 80%-20% train-test split was used.

## 4. Results

The table below presents the performance of multilabel classification algorithms evaluated across the selected datasets. For binary relevance and classifier chains, we used Logistic Regression as the base classifier. For RAKEL, we took subset size to be 3.

Table 1: Results for Drug Consumption Dataset

| <b>Model</b>                     | <i>Subset Accuracy</i> | <i>Hamming Loss</i> | <i>Micro F1</i> | <i>Macro F1</i> | <i>ROC-AUC</i> |
|----------------------------------|------------------------|---------------------|-----------------|-----------------|----------------|
| <i>Binary Relevance</i>          | 0.3846                 | 0.2347              | 0.7093          | 0.6712          | 0.7268         |
| <i>Classifier Chains</i>         | <b>0.3978</b>          | 0.2400              | <b>0.7206</b>   | <b>0.6943</b>   | <b>0.7401</b>  |
| <i>Multi-label Decision Tree</i> | 0.3660                 | <b>0.2745</b>       | 0.6693          | 0.6407          | 0.6970         |
| <i>RAkEL (k=3)</i>               | 0.2355                 | 0.2411              | 0.6980          | 0.5712          | 0.6348         |

Table 2: Results for Enzyme Substrate (Chemical)

| <b>Model</b>                     | <i>Subset Accuracy</i> | <i>Hamming Loss</i> | <i>Micro F1</i> | <i>Macro F1</i> | <i>ROC-AUC</i> |
|----------------------------------|------------------------|---------------------|-----------------|-----------------|----------------|
| <i>Binary Relevance</i>          | 0.2355                 | 0.2411              | <b>0.6980</b>   | 0.5712          | <b>0.6348</b>  |
| <i>Classifier Chains</i>         | 0.2596                 | 0.2387              | 0.6977          | <b>0.5646</b>   | 0.6337         |
| <i>Multi-label Decision Tree</i> | <b>0.2740</b>          | 0.2556              | 0.6741          | 0.5422          | 0.6266         |
| <i>RAkEL (k=3)</i>               | 0.2355                 | <b>0.2652</b>       | 0.6699          | 0.5451          | 0.6274         |

Table 3: Results for Enzyme Substrate (Structural)

| <b>Model</b>                     | <i>Subset Accuracy</i> | <i>Hamming Loss</i> | <i>Micro F1</i> | <i>Macro F1</i> | <i>ROC-AUC</i> |
|----------------------------------|------------------------|---------------------|-----------------|-----------------|----------------|
| <i>Binary Relevance</i>          | 0.1538                 | 0.3060              | 0.5430          | 0.4669          | <b>0.5958</b>  |
| <i>Classifier Chains</i>         | <b>0.1586</b>          | <b>0.3076</b>       | <b>0.5439</b>   | <b>0.4679</b>   | 0.5949         |
| <i>Multi-label Decision Tree</i> | 0.1394                 | 0.2972              | 0.5212          | 0.3969          | 0.5604         |
| <i>RAkEL (k=3)</i>               | 0.1346                 | 0.3004              | 0.5247          | 0.4210          | 0.5779         |

Table 4: Results for Enzyme Substrate (Connectivity)

| <b>Model</b>                     | <i>Subset Accuracy</i> | <i>Hamming Loss</i> | <i>Micro F1</i> | <i>Macro F1</i> | <i>ROC-AUC</i> |
|----------------------------------|------------------------|---------------------|-----------------|-----------------|----------------|
| <i>Binary Relevance</i>          | 0.1009                 | <b>0.3076</b>       | 0.5339          | 0.4641          | 0.5925         |
| <i>Classifier Chains</i>         | 0.1009                 | 0.3036              | 0.5372          | <b>0.4806</b>   | 0.6040         |
| <i>Multi-label Decision Tree</i> | <b>0.2019</b>          | 0.2652              | <b>0.5761</b>   | 0.4656          | <b>0.6047</b>  |
| <i>RAkEL (k=3)</i>               | 0.1201                 | 0.3125              | 0.5301          | 0.4353          | 0.5781         |

Classifier Chains achieved the best overall performance on the Drug Consumption dataset with the highest Subset Accuracy, Micro F1-score, Macro F1-score, and ROC-AUC values. This indicates that incorporating label dependencies improved prediction quality significantly. Binary Relevance also showed competitive performance despite treating labels independently. The Multi-label Decision Tree produced comparatively higher Hamming Loss and lower F1-scores, suggesting difficulty in modelling complex label relationships. RAKEL showed moderate performance but did not outperform simpler methods on this dataset.

For the Chemical Enzyme Substrate dataset, the Multi-label Decision Tree achieved the highest Subset Accuracy, indicating improved exact-match prediction capability. However, Binary Relevance and Classifier Chains obtained higher Micro and Macro F1-scores, demonstrating better overall multilabel prediction quality. Classifier Chains also achieved the lowest Hamming Loss among the evaluated methods. RAKEL again showed comparatively lower performance, particularly in Hamming Loss and F1-score metrics.

The Structural Enzyme Substrate dataset proved more challenging than the previous datasets, as indicated by the lower overall performance across all models. Classifier Chains achieved slightly higher Subset Accuracy and F1-scores compared to Binary Relevance, suggesting modest benefits from modelling label dependencies. The Multi-label Decision Tree achieved the lowest Hamming Loss but produced weaker F1 and ROC-AUC scores. RAKEL demonstrated balanced but moderate performance across metrics.

For the Connectivity dataset, the Multi-label Decision Tree significantly outperformed the other methods in terms of Subset Accuracy and Micro F1-score while also producing the lowest Hamming Loss. This suggests that hierarchical partitioning of the feature space was particularly effective for this dataset. Classifier Chains achieved the highest Macro F1-score and competitive ROC-AUC performance, indicating better balanced prediction across labels. RAKEL showed the weakest overall performance on this dataset.

## 5. Conclusions

The experimental results demonstrate that multilabel classifier performance varies considerably depending on dataset characteristics and feature representations.

Classifier Chains consistently achieved strong overall performance across multiple datasets, particularly in terms of Micro F1-score, Macro F1-score, and ROC-AUC. This highlights the

importance of modelling label dependencies in multilabel learning tasks. Binary Relevance provided competitive baseline performance despite its simpler independent-label assumption. Its relatively strong results suggest that certain datasets may contain weaker label correlations.

Multi-label Decision Trees achieved superior exact-match prediction performance on some datasets, particularly the Connectivity dataset, while also producing lower Hamming Loss values in certain cases. However, their performance was less consistent across all datasets. RAKEL demonstrated moderate performance overall but generally did not outperform Binary Relevance or Classifier Chains. This may indicate that the random label subset ensemble strategy was less effective for the evaluated datasets.

The study demonstrates that multilabel classification performance is highly dependent on both the learning strategy and the underlying dataset structure. Methods that explicitly capture label correlations generally achieve better predictive performance than approaches treating labels independently.

Future work may explore advanced deep learning-based multilabel architectures, graph neural networks for label dependency modelling, attention mechanisms, transformer-based methods, and hybrid ensemble techniques for improving multilabel prediction performance on complex high-dimensional datasets.

## 6. References:

[1] Drug Consumption Dataset

URL (as of 20-04-26): <https://www.kaggle.com/datasets/mexwell/drug-consumption-classification>

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